

7. Components of 70S ribosomes are

- (a) 40S and 30S (b) 50S and 30S
(c) 50S and 20S (d) 40S and 40S

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (b)

Ribosomes are primary cellular component that are involved in Protein Synthesis. Eukaryotic cells consist of 70S and 80S Ribosomes and Prokaryotes consist of 70S Ribosomes. 70S Ribosomes consist of the smaller subunit 30S and larger is 50S.

8. Which of the following is found both in the cells of a mango plant and a monkey?

- (a) Chloroplast (b) Centriole
(c) Cell wall (d) Cell membrane
(e) None of the above / More than one of the above

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Ans. (d)

The cell membrane, also called the plasma membrane, is found in all cells (plant and Animal) and separates the interior of the cell from the outside environment. The cell membrane consists of a lipid bilayer and protein that is semipermeable. The cell membrane regulates the transport of materials entering and exiting the cell.

9. What is the expanded form of the term 'mRNA' that has been widely discussed since the beginning of the pandemic?

- (a) Messenger Ribonucleic Acid
(b) Mutant Ribonucleic Acid
(c) Modified Ribonucleic Acid
(d) Mnemonic Ribonucleic Acid

69th B.P.S.C. (Pre) 2023

Ans. (a)

mRNA stands for Messenger Ribonucleic Acid and is the single stranded molecule that carries the instructions to make proteins.

Heredity

1. The process where the characteristics are transmitted from parents to offsprings is called

- (a) variation (b) heredity
(c) gene (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Heredity refers to the genetic heritage passed down by our biological parents. It is a process where the characteristics are transmitted from parents to off springs. It's why we look like them! More specifically, it is the transmission of traits from one generation to the next.

2. Mendel's principle of inheritance is based on-

- (a) Vegetative reproduction
(b) Asexual reproduction
(c) Sexual reproduction
(d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Mendel's principle of inheritance is based on sexual reproduction.

3. Which of the following is responsible for controlling hereditary characteristics of the living cells?

- (a) Enzyme (b) Hormone
(c) RNA (d) DNA
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (d)

DNA is responsible for storing and transferring genetic information while RNA directly codes for amino acids and acts as a messenger between DNA and ribosomes to make proteins. However, studies showed that RNA replaces DNA in some viruses like Tobacco Mosaic virus, bacteriophage, etc. So, not every organism has DNA as their hereditary material, some have RNA. But, DNA is the predominant one and in the majority of higher level organisms, RNA plays the role of a messenger.

4. Sexual reproduction causes genetic variation because of -

- (a) Blending of genes (b) Chromosomal changes
(c) Shuffling of genes (d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Sexual reproduction involves two parent cells; each parent gives some of its traits (characteristics) to the offspring. Sexual reproduction causes, genetic variation because of blending of genes, chromosomal changes, shuffling of genes etc.

5. Who discovered DNA?

- (a) James Watson and Francis Crick
(b) Gregor Mendel
(c) Johannsen
(d) Har Gobind Khorana
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (e)

Friedrich Miescher first identified and isolated (discovered) DNA in 1869. Double helix model of DNA was given by James Watson and Francis Crick in 1953.